$$F(x, y, z) = 2x^2 - y^2 - z^2$$

Let $\vec{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$F(\vec{r}) = \vec{r}^{\,T} M \vec{r}$$

where $M = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$